

LET THE NEURONS DIE: Exploiting ReLU-Induced Model Degradation

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INTRODUCTION

As machine learning becomes more prevalent for critical applications, the underlying model architecture (e.g., activation function) poses vulnerability to a variety of attacks. **ReLU** is a popular activation function for its simplicity, low computational cost, and performance. However, ReLU can suffer from **dead neurons** which adversaries can abuse.

THREAT MODELS

Point of Attack: Prior to training

Attack Preparation Phase:

White-box access to victim model architecture, dataset and initial weights

Attack Execution Phase:

- **DOA:** requires NO new data injection
- **IG-SKA & IG-DOA:** requires new poisoned data injection

METHODOLOGY

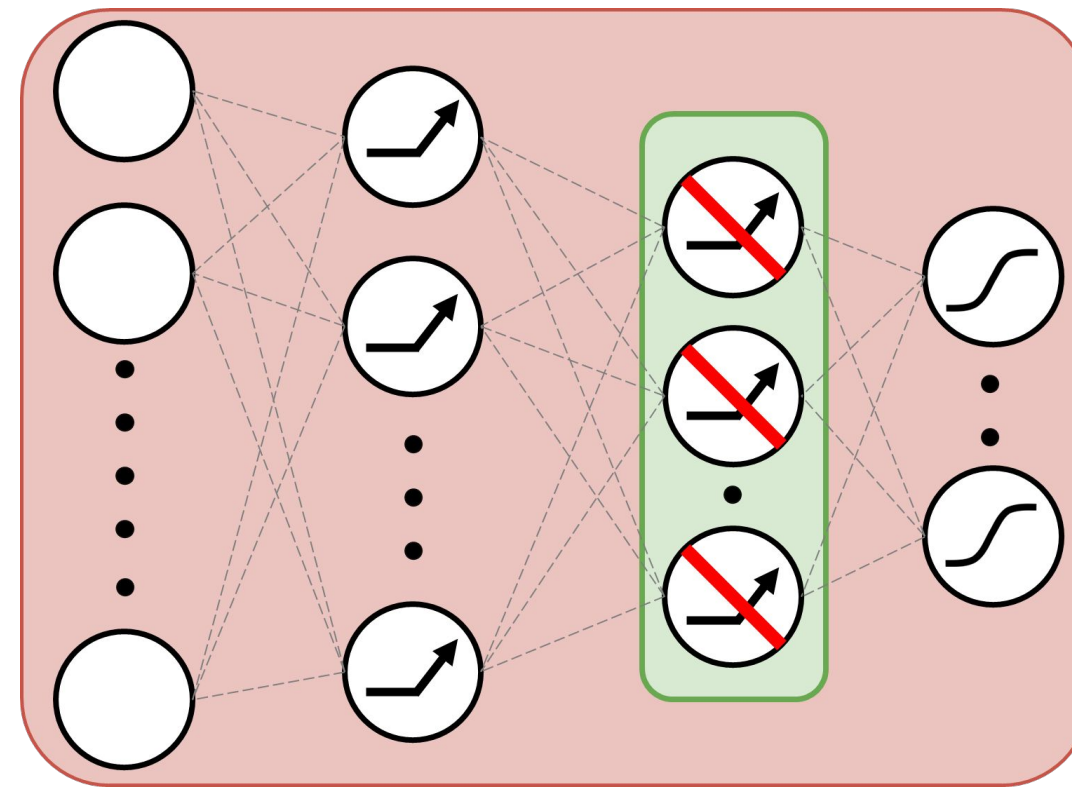
Dynamic Data Ordering Attack (DOA)

Input: Victim model's initial weights, $\mathbf{W}_i$

Idea: A target layer ℓ with ReLU activations is targeted to minimize weights

1. Train data points (1...n) individually from $\mathbf{W}_i$
2. Calculate $\text{score}_k = \sum \mathbf{W}_{k\ell}$ = Sum of weights for layer ℓ after training on point $\mathbf{k}$
3. Add data point $\mathbf{k}$ with lowest score to candidate list
4. Repeat 1-3 starting from $\mathbf{W}_k$

Output: Candidate list of data ordered points negatively biasing layer ℓ



Multi-Layer Data Poisoning Attacks

Frontend Attack

Option 1: SKA

Input: Victim model's initial weights, $\mathbf{W}_i$

Idea: e.g., for every 2 consecutive layers with ReLU, weights $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{B} \in \mathbb{R}^{\ell \times m}$ and biases $\mathbf{a} \in \mathbb{R}^m$, $\mathbf{b} \in \mathbb{R}^\ell$, $y = \text{ReLU}(\mathbf{B} \text{ReLU}(\mathbf{A}\mathbf{x} + \mathbf{a}) + \mathbf{b})$, we concentrate +ve components and cross $\mathbf{A}$ & $\mathbf{B}$, aiming to annihilate $\mathbf{W}$

$$\text{ReLU}\left(\begin{bmatrix} \text{matrix A} \\ \vdots \end{bmatrix} \mathbf{x} + \mathbf{a}\right) = \begin{bmatrix} \vdots \\ \vdots \end{bmatrix} \quad \text{ReLU}\left(\begin{bmatrix} \text{matrix B} \\ \vdots \end{bmatrix} \begin{bmatrix} \vdots \\ \vdots \end{bmatrix} + \mathbf{b}\right) = \begin{bmatrix} \vdots \\ \vdots \end{bmatrix} = \mathbf{y}$$

Output: Harmful weights impeding training, $\mathbf{W}_t$

Option 2: DOA

Output: Weights from model trained with ordered dataset, $\mathbf{W}_t$

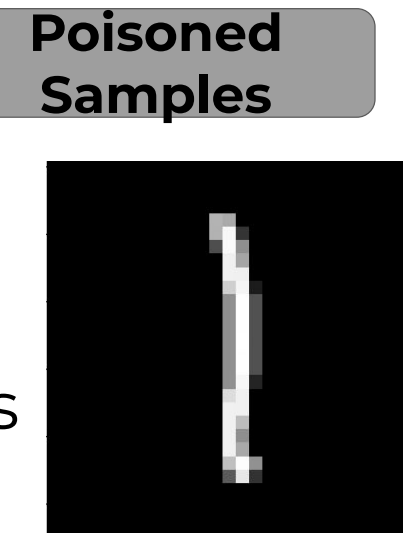
Frontend Attack

Option 1: SKA
Option 2: DOA

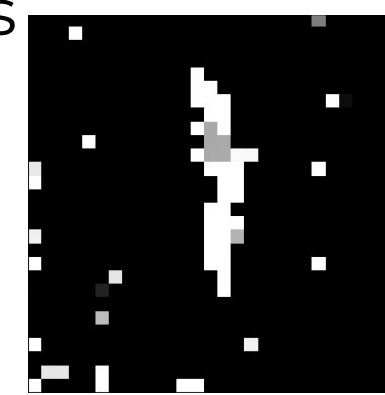
Model Weights

Backend Attack

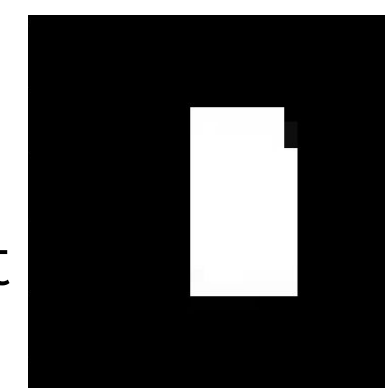
IG



TRUE



IG-SKA



IG-DOA

IG

Original Inverting Gradient (IG) Setting:

- In **Federated Learning**, each user trains a model on a server with **private** local data. They receive current weights from the server and share parameter updates (gradients)
- Original Goal: Training data reconstruction from gradient information (privacy infringement)

IG-SKA & IG-DOA

Input: Victim model & $\mathbf{W}_t$

Idea: Initialize FL model with $\mathbf{W}_t$ and **repurpose** IG to construct poisoned input

Output: Poisoned data sample(s) for every class

DISCUSSION

Dynamic Data Ordering Attack

- + Difficult to detect
- + Minimally invasive as only a small subset of data points need ordering
- Smaller performance degradation
- Computationally expensive

Multi-Layer Data Poisoning Attacks

- + Aggressive performance degradation
- + Can craft many adversaries
- + More obvious to detect

IG-SKA

- + Has higher potential in the best case
- Mathematically derived, and deviating from exact values limits potential gain

IG-DOA

- + More consistent in average cases
- Computationally expensive
- Easier to detect

FUTURE WORK

Attack Effectiveness:

- Study the correlation of #samples ordered/poisoned vs accuracy drop
- Maintain or exceed attack quality with fewer ordered/poisoned samples

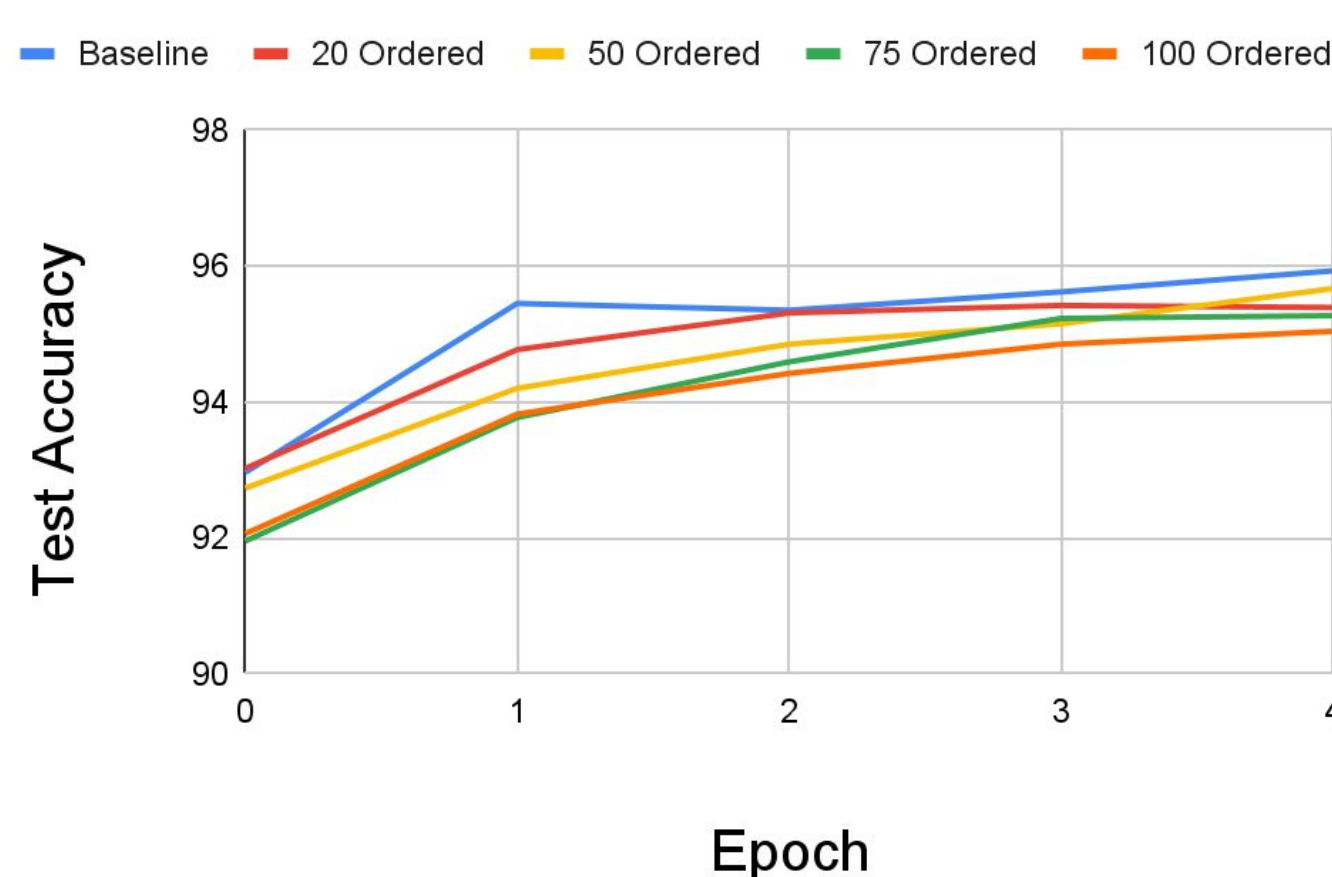
Attack Extensibility & Scalability:

- Test on more model architectures (e.g., CNNs, ResNet, etc.)
- Test with more datasets (e.g., CIFAR, ImageNet, etc.)
- Translate to other modalities (e.g., text)

RESULTS

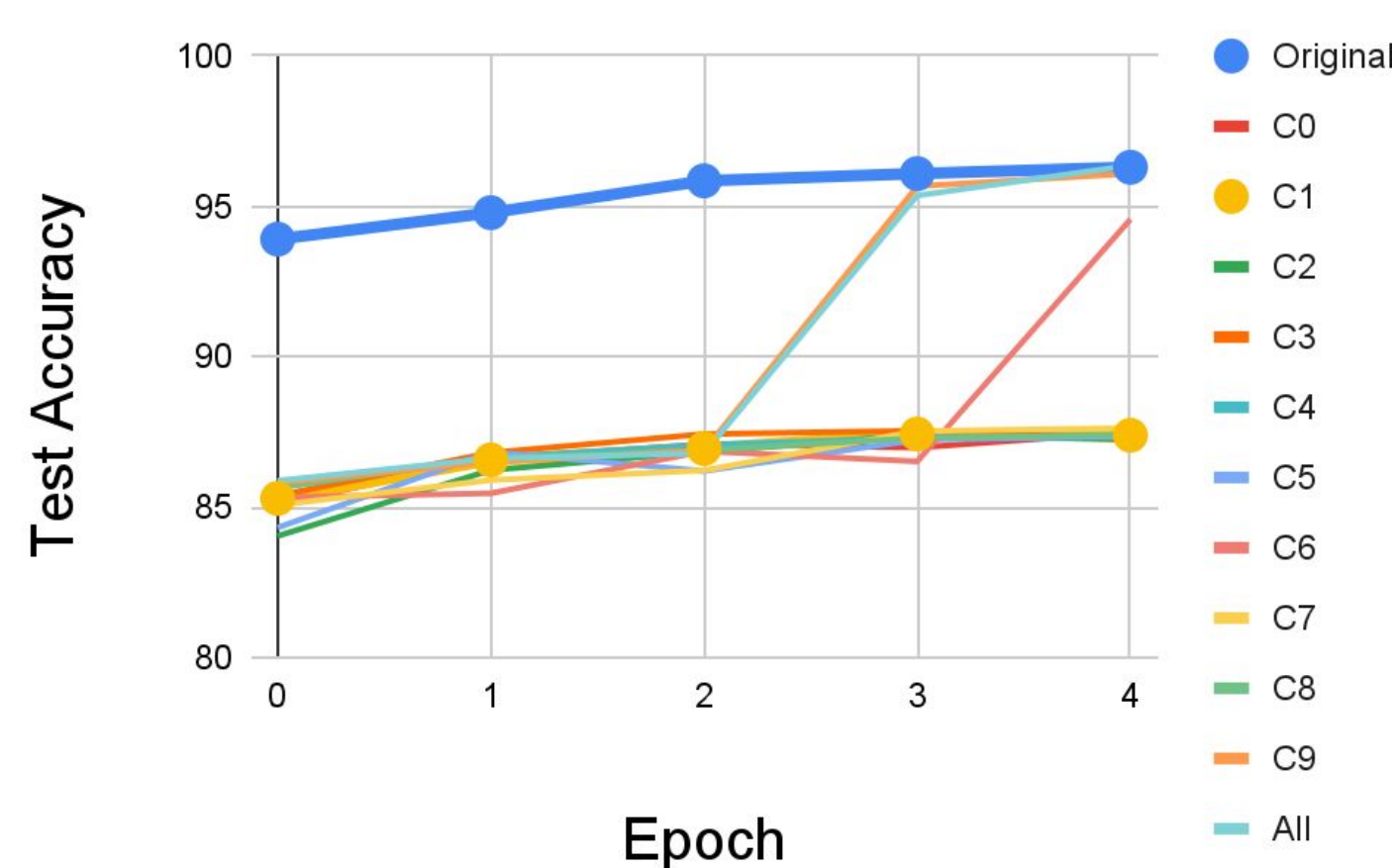
Note: All the experiments were run on **MNIST** dataset

DOA



Ordered indicates the accuracy on the test dataset at each epoch when ordering # of data points during training

IG - SKA



C# indicates the accuracy on the test dataset at each epoch when adding 200 poisoned samples reconstructed from Class # into the training dataset **shuffled** (All indicates poisoning all classes)

IG - DOA

